Technical Document #219: Data Engineering - Comprehensive Overview

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ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Data governance establishes policies and procedures for managing data assets.
- Data pipelines automate the movement and transformation of data between systems.
- Data lineage tracks the origin and movement of data through systems.